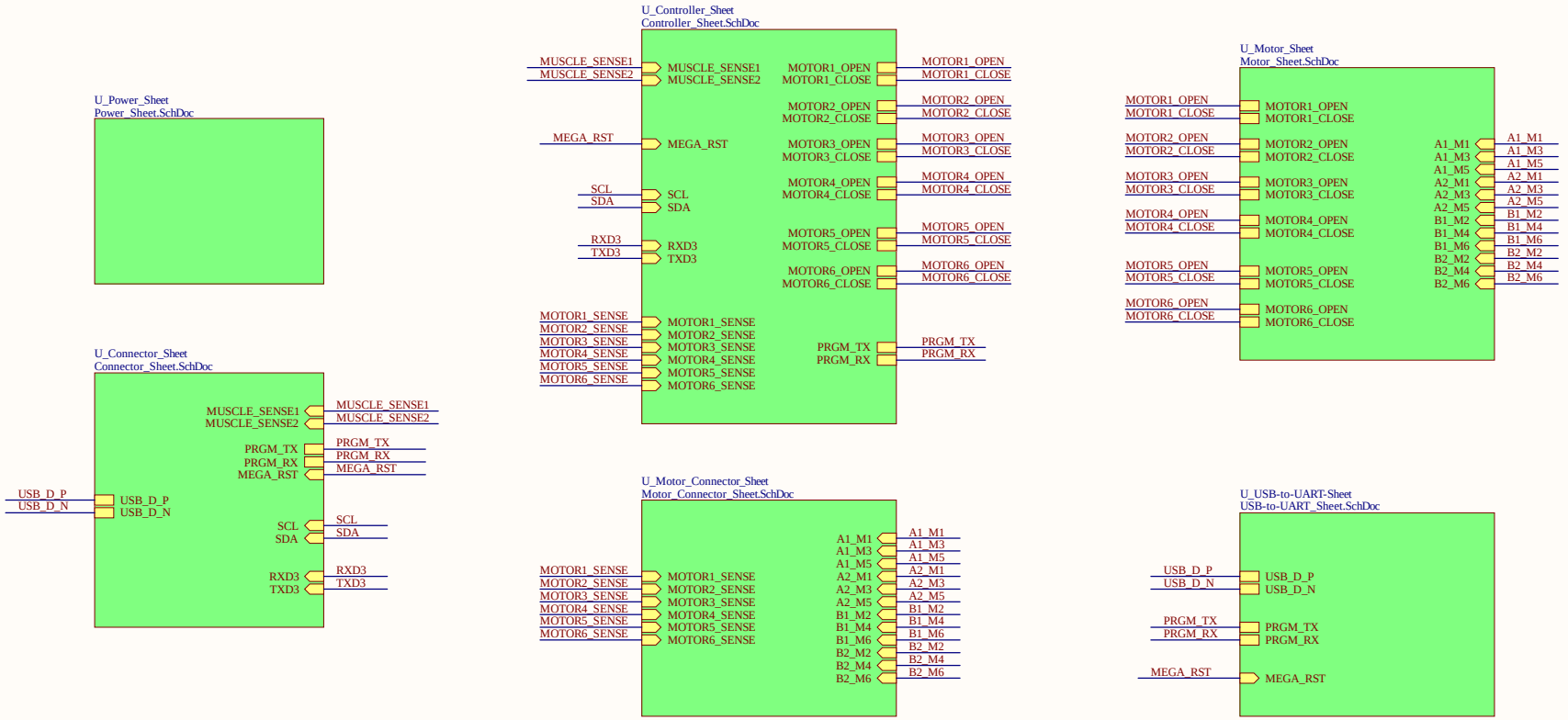
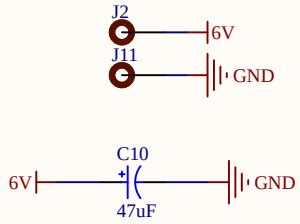
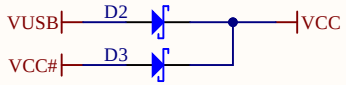

U_Power_Sheet
Power_Sheet.SchDoc
U_Connector_Sheet
Connector_Sheet.SchDoc
MUSCLE_SENSE1
MUSCLE_SENSE2
PRGM_TX
PRGM_RX
MEGA_RST
USB_D_P
USB_D_N
SCL
SDA
RXD3
TXD3
U_Controller_Sheet
Controller_Sheet.SchDoc
MOTOR1_OPEN
MOTOR1_CLOSE
MOTOR2_OPEN
MOTOR2_CLOSE
MOTOR3_OPEN
MOTOR3_CLOSE
MOTOR4_OPEN
MOTOR4_CLOSE
MOTOR5_OPEN
MOTOR5_CLOSE
MOTOR6_OPEN
MOTOR6_CLOSE
MOTOR1_SENSE
MOTOR2_SENSE
MOTOR3_SENSE
MOTOR4_SENSE
MOTOR5_SENSE
MOTOR6_SENSE
U_Motor_Sheet
Motor_Sheet.SchDoc
A1_M1
A1_M3
A1_M5
A2_M1
A2_M3
A2_M5
B1_M2
B1_M4
B1_M6
B2_M2
B2_M4
B2_M6
U_Motor_Connector_Sheet
Motor_Connector_Sheet.SchDoc
U_USB-to-UART-Sheet
USB-to-UART_Sheet.SchDoc

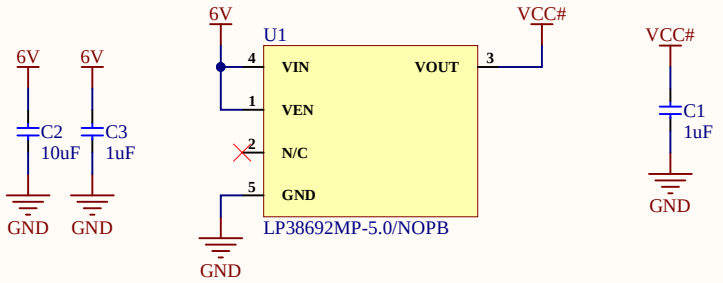
POWER INPUTS

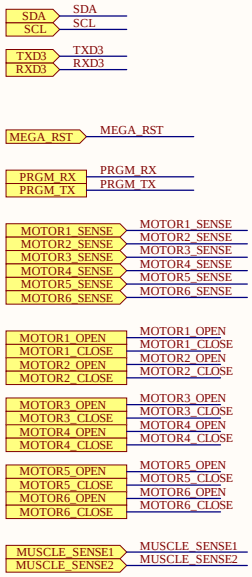


DIODE ORING

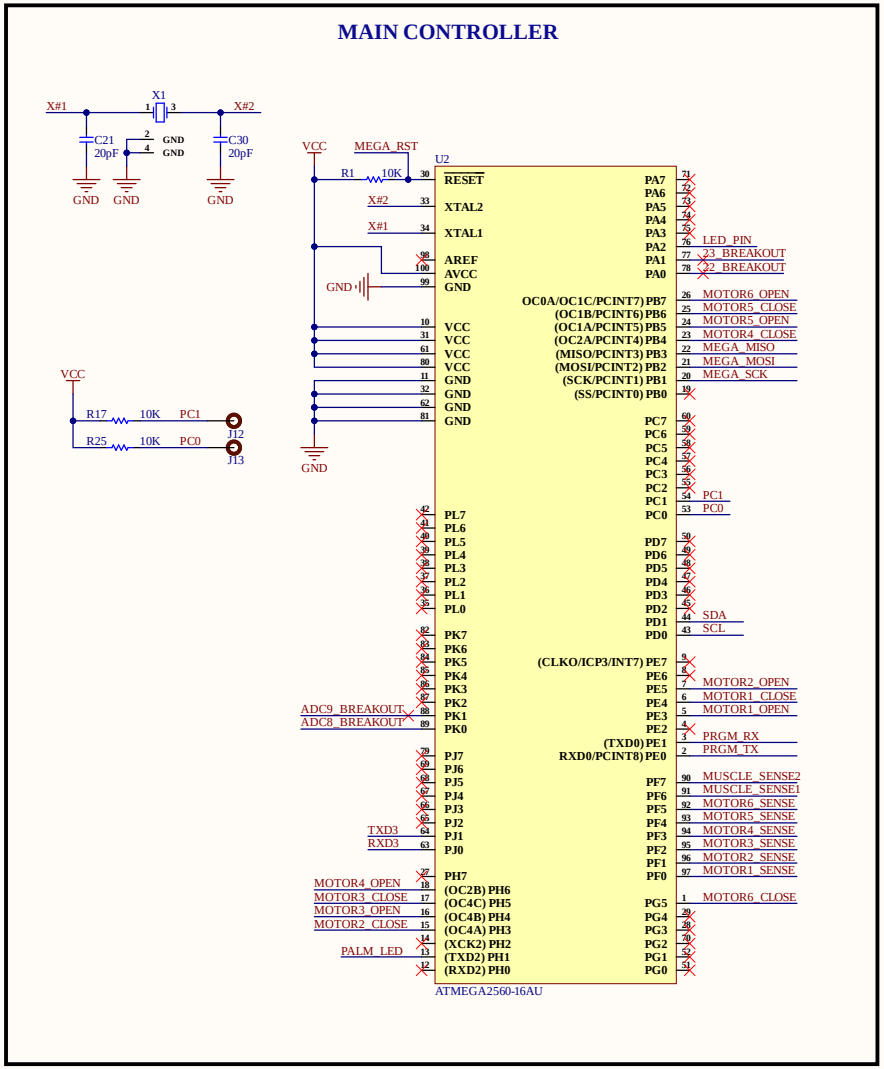


6V TO 5V CONVETER

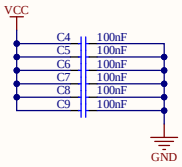




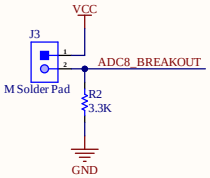
MAIN CONTROLLER



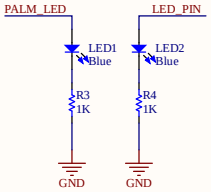
DECOUPLING CAPACITORS



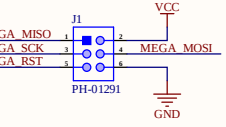
SENSOR CONNECTION



LED'S



SPI



B2_M6 B2_M6
B1_M6 B1_M6

A2_M5 A2_M5
A1_M5 A1_M5

B2_M4 B2_M4
B1_M4 B1_M4

A2_M3 A2_M3
A1_M3 A1_M3

B2_M2 B2_M2
B1_M2 B1_M2

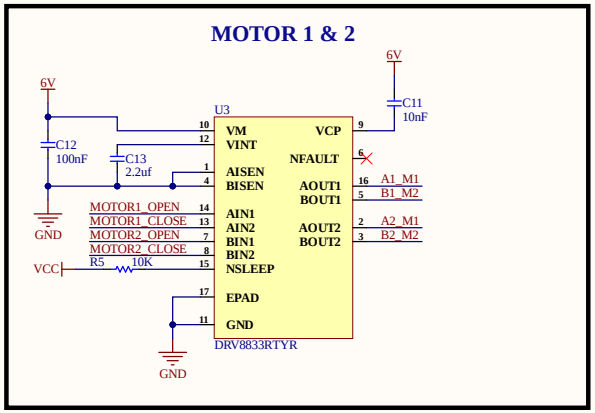
A2_M1 A2_M1
A1_M1 A1_M1

MOTOR1_OPEN MOTOR1_OPEN
MOTOR1_CLOSE MOTOR1_CLOSE
MOTOR2_OPEN MOTOR2_OPEN
MOTOR2_CLOSE MOTOR2_CLOSE

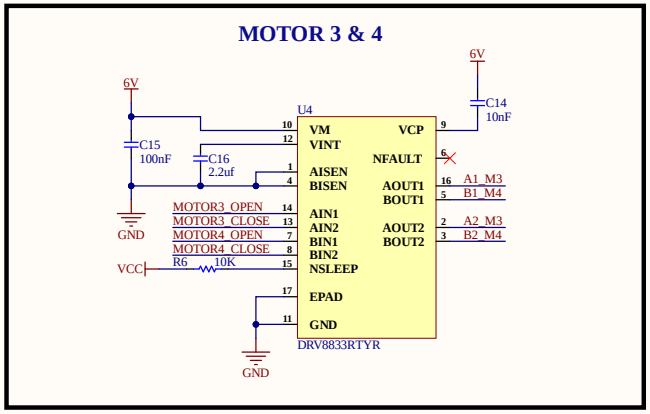
MOTOR3_OPEN MOTOR3_OPEN
MOTOR3_CLOSE MOTOR3_CLOSE
MOTOR4_OPEN MOTOR4_OPEN
MOTOR4_CLOSE MOTOR4_CLOSE

MOTOR5_OPEN MOTOR5_OPEN
MOTOR5_CLOSE MOTOR5_CLOSE
MOTOR6_OPEN MOTOR6_OPEN
MOTOR6_CLOSE MOTOR6_CLOSE

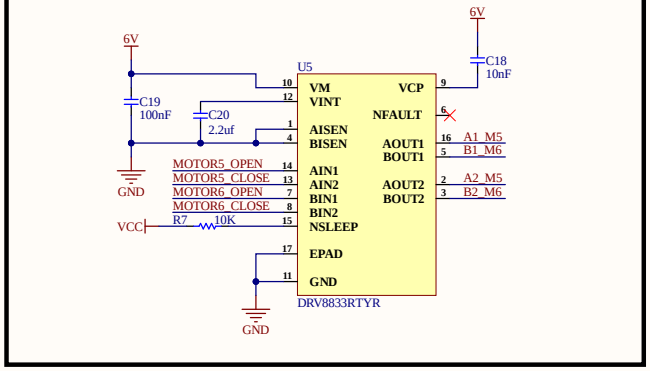
MOTOR 1 & 2

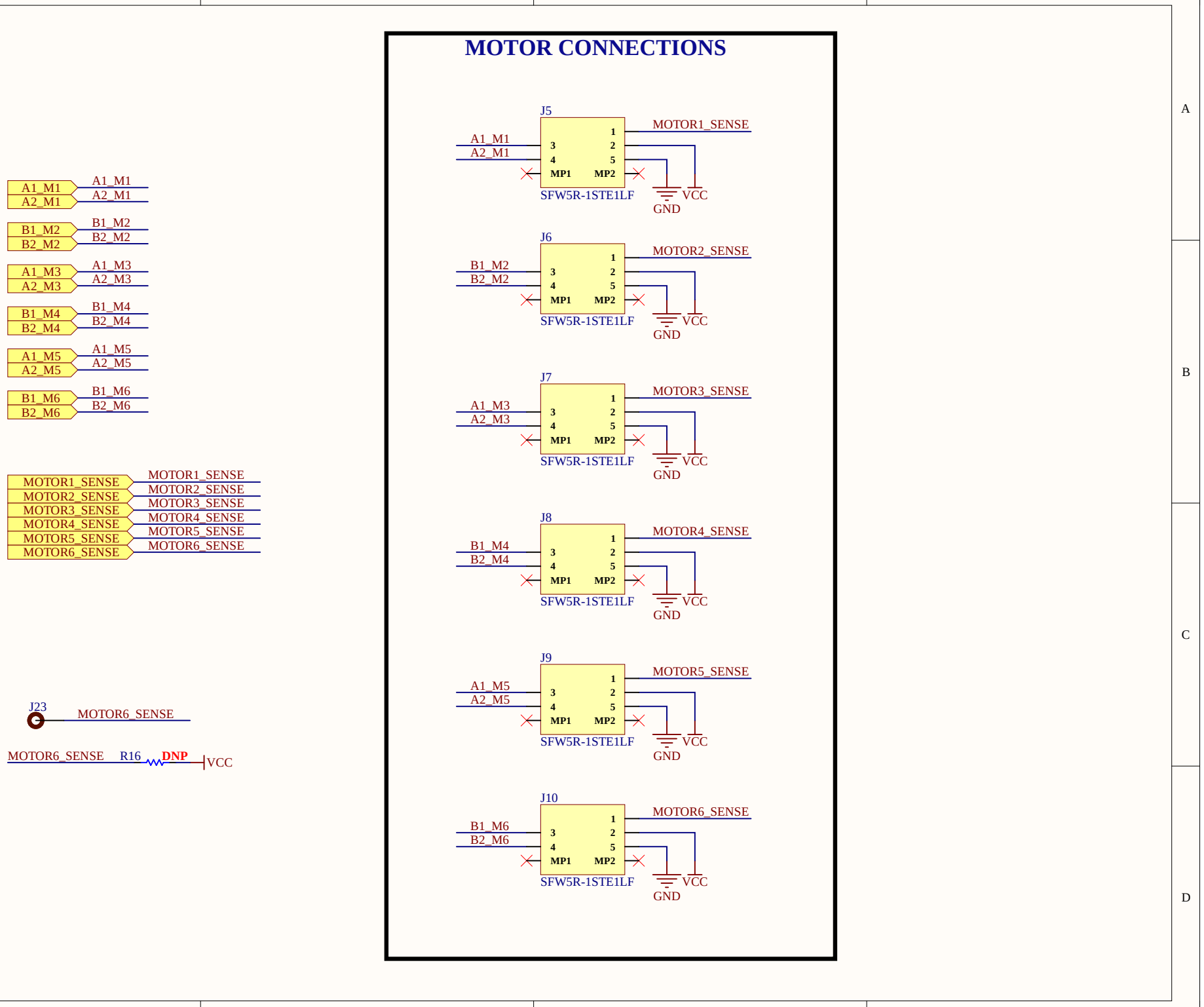


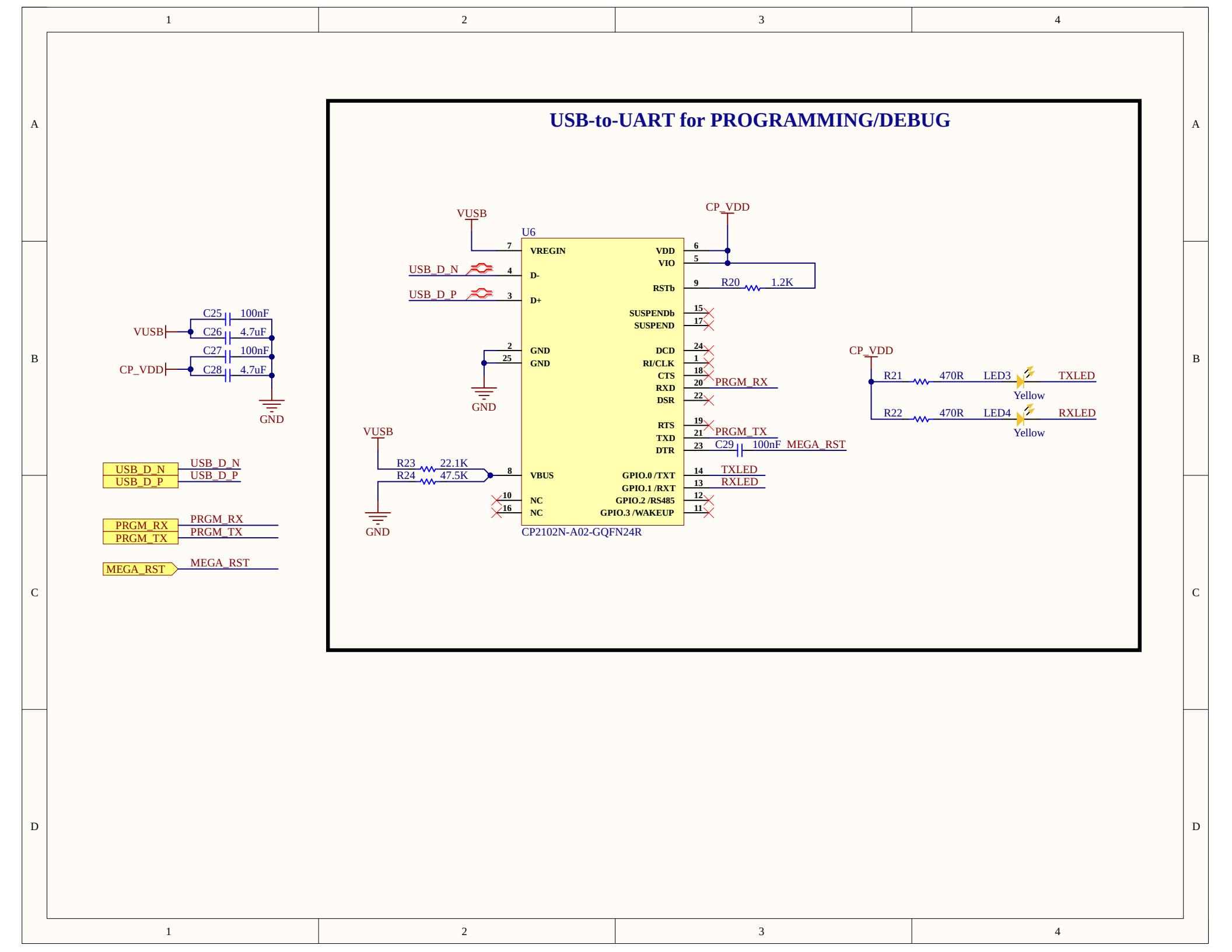
MOTOR 3 & 4



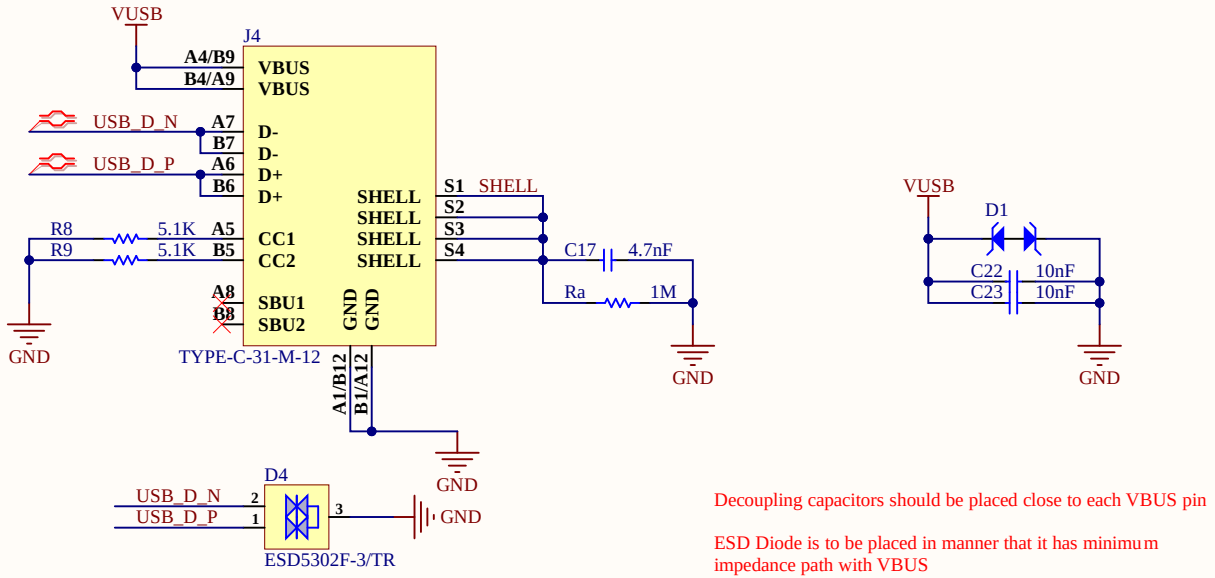
MOTOR 5 & 6




MOTOR CONNECTIONS
J5
A1_M1
A2_M1
MOTOR1_SENSE
SFW5R-1STE1LF
VCC
GND
J6
B1_M2
B2_M2
MOTOR2_SENSE
J7
A1_M3
A2_M3
MOTOR3_SENSE
J8
B1_M4
B2_M4
MOTOR4_SENSE
J9
A1_M5
A2_M5
MOTOR5_SENSE
J10
B1_M6
B2_M6
MOTOR6_SENSE
J23
MOTOR6_SENSE
R16
DNP


USB-to-UART for PROGRAMMING/DEBUG
VUSB
C25 100nF
C26 4.7uF
C27 100nF
CP_VDD
C28 4.7uF
GND
USB_D_N
USB_D_P
PRGM_RX
PRGM_TX
MEGA_RST
U6
VREGIN
D-
D+
GND
GND
VBUS
NC
NC
VDD
VIO
RSTb
SUSPENDb
SUSPEND
DCD
RI/CLK
CTS
RXD
DSR
RTS
TXD
DTR
GPIO.0 /TXT
GPIO.1 /RXT
GPIO.2 /RS485
GPIO.3 /WAKEUP
CP2102N-A02-GQFN24R
R20 1.2K
R23 22.1K
R24 47.5K
C29 100nF
R21 470R
LED3
Yellow
R22 470R
LED4
Yellow
TXLED
RXLED

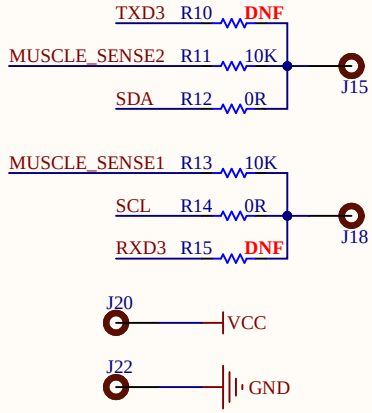
USB-C CONNECTOR

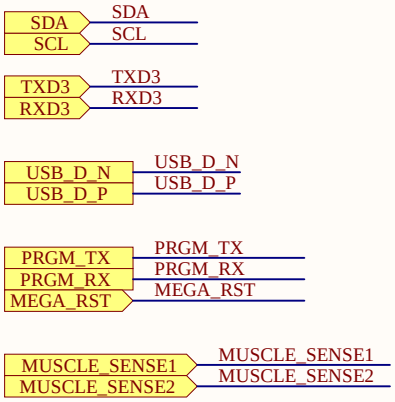


Decoupling capacitors should be placed close to each VBUS pin

ESD Diode is to be placed in manner that it has minimum impedance path with VBUS

AUDIO CONNECTION





USB CONNECTION

